

Qualification Pack



Water Hyacinth (Value added Products) Entrepreneur

QP Code: SGJ/Q4105

Version: 1.0

NSQF Level: 4.0

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SGJ/Q4105: Water Hyacinth (Value added Products) Entrepreneur

Brief Job Description

The purpose of this qualification is to increase livelihood opportunities for the unskilled population and promote entrepreneurial skills through systematic training on the collection, processing, value-addition and marketing of products made from water hyacinth. By turning a challenging weed into a valuable resource, not only can environmental harm be mitigated, but local economies can also be bolstered through the creation of sustainable livelihoods.

Personal Attributes

The individual should be eco-conscious, creative, and self-motivated, with a strong interest in sustainability and local entrepreneurship. They must be skilled in hands-on work, willing to learn, and adaptable to new techniques. Good communication skills, attention to detail, problem-solving ability, and a basic understanding of business and marketing are essential for success in this role. A sense of responsibility towards the community and the environment is also important.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4118: Introduce course collection, cleaning and processing of water hyacinth.](#)
2. [SGJ/N4119: Utilise water Hyacinth for Pulp Processing and Paper making](#)
3. [SGJ/N4120: Utilise water Hyacinth for Weaving and handicraft making.](#)
4. [SGJ/N4121: Utilise water Hyacinth for Mushroom Cultivation](#)
5. [SGJ/N4122: Utilise water Hyacinth for Briquette production.](#)
6. [SGJ/N4123: Utilise water Hyacinth for Biogas production.](#)
7. [SGJ/N4124: Finishing and Marketing of Products](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Forestry

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Occupation	Entrepreneur
Country	India
NSQF Level	4.0
Credits	13
Aligned to NCO/ISCO/ISIC Code	7319 (Handicraft Workers not classified elsewhere)
Minimum Educational Qualification & Experience	10th grade pass with 2 Years of experience of relevant experience in handicraft industries OR 12th grade Pass (or equivalent) with NA of experience NA OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience of relevant experience in any handicraft industries OR Previous relevant Qualification of NSQF Level (3.5) with 1-2 Years of experience 1.5-years of relevant experience in any handicraft industries
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-04-ES-04260-2025-V1-SCGJ
NQR Version	1

Remarks:

Total 390 hours (Theory: 140 hours with employability skills, Practical: 160 hours, OJT: 90 hours)

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SGJ/N4118: Introduce course collection, cleaning and processing of water hyacinth.

Description

The NOS introduces learners to the concept of water hyacinth as an invasive aquatic weed and its potential as a valuable resource. The learner gains knowledge about identifying water hyacinth in natural water bodies, understanding its environmental implications, and learning proper techniques for collection, cleaning, and post-harvest drying.

Scope

The scope covers the following :

- This scope covers the foundational understanding and basic skills required for identifying, collecting, cleaning, and preserving water hyacinth as a raw material for various value-added applications. It includes knowledge about the nature and impact of the water hyacinth plant, safe and sustainable harvesting methods, and the initial steps involved in preparing the plant for further use in handicrafts, paper, biogas, briquette, or mushroom cultivation processes.

Elements and Performance Criteria

Introduction to Water hyacinth, source of water hyacinth and its types etc.

To be competent, the user/individual on the job must be able to:

- PC1.** Describe about the problems of water hyacinth growing in waterbodies.
- PC2.** Discuss how it is a problem for fishing community
- PC3.** Discuss how it is economically viable
- PC4.** Describe the concept of invasive species
- PC5.** Discuss the invasive weed water hyacinth
- PC6.** Discuss the management of the invasive weed species
- PC7.** Discuss the need for the eradication procedures of invasive weed
- PC8.** Demonstrate and show water hyacinth plant in various water bodies, the stem of the plant and various products made out of it.
- PC9.** Demonstrate a range of cognitive and practical skills required to identify polluted areas.
- PC10.** Demonstrate polluted water bodies, identify different weed and extraction process.

Raw Material Preparation and Preservation of Water Hyacinth

To be competent, the user/individual on the job must be able to:

- PC11.** Understand the invasive weed water hyacinth.
- PC12.** Understand the procedures for collecting water hyacinth.
- PC13.** Discuss the techniques that are involved to descend into water body and harvest the plant.
- PC14.** Discuss how to cut leaves and root off the stems and laid out to dry in sunlight
- PC15.** Discuss post-harvest processing of water hyacinth like cleaning, drying and managing raw material.
- PC16.** Demonstrate how the plant will be cut and brought out to be taken away for drying.

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PC17. Demonstrate operating different types of machines/tools.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Concept of invasive species and their impact on the ecosystem.
- KU2.** Characteristics and growth pattern of water hyacinth.
- KU3.** Environmental and economic problems caused by water hyacinth.
- KU4.** Benefits of removing and utilizing water hyacinth.
- KU5.** Procedures for collecting water hyacinth from different water bodies.
- KU6.** Methods of safe cutting and handling of the plant.
- KU7.** Storage and preservation methods for raw water hyacinth material.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate clearly with team members during field operations.
- GS2.** Follow verbal instructions for collection, cleaning, and safety.
- GS3.** Read and understand standard operating procedures.
- GS4.** Record details of raw material collected and processed.
- GS5.** Work as part of a team in harvesting and handling activities.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Water hyacinth, source of water hyacinth and its types etc.</i>	9	15	-	-
PC1. Describe about the problems of water hyacinth growing in waterbodies.	2	-	-	-
PC2. Discuss how it is a problem for fishing community	2	-	-	-
PC3. Discuss how it is economically viable	2	-	-	-
PC4. Describe the concept of invasive species	2	-	-	-
PC5. Discuss the invasive weed water hyacinth	1	-	-	-
PC6. Discuss the management of the invasive weed species	-	5	-	-
PC7. Discuss the need for the eradication procedures of invasive weed	-	5	-	-
PC8. Demonstrate and show water hyacinth plant in various water bodies, the stem of the plant and various products made out of it.	-	5	-	-
PC9. Demonstrate a range of cognitive and practical skills required to identify polluted areas.	-	-	-	-
PC10. Demonstrate polluted water bodies, identify different weed and extraction process.	-	-	-	-
<i>Raw Material Preparation and Preservation of Water Hyacinth</i>	11	15	-	-
PC11. Understand the invasive weed water hyacinth.	2	-	-	-
PC12. Understand the procedures for collecting water hyacinth.	3	-	-	-
PC13. Discuss the techniques that are involved to descend into water body and harvest the plant.	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Discuss how to cut leaves and root off the stems and laid out to dry in sunlight	3	-	-	-
PC15. Discuss post-harvest processing of water hyacinth like cleaning, drying and managing raw material.	-	5	-	-
PC16. Demonstrate how the plant will be cut and brought out to be taken away for drying.	-	5	-	-
PC17. Demonstrate operating different types of machines/tools.	-	5	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4118
NOS Name	Introduce course collection, cleaning and processing of water hyacinth.
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4.0
Credits	1.0
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N4119: Utilise water Hyacinth for Pulp Processing and Paper making

Description

This NOS enables individuals to transform water hyacinth into usable pulp and produce handmade paper or molded pulp items. The learners are trained in pulping techniques, preparation of moulds and deckles, and the sheet-forming process. They also learn to carry out quality checks, smooth and trim finished paper, and apply basic binding and decorative methods.

Scope

The scope covers the following :

- The Scope covers the skills and knowledge required to process water hyacinth into pulp and utilize it for handmade paper and pulp-based product creation. It includes pulping techniques, mould and deckle-based papermaking processes, and finishing methods such as trimming, embossing, and binding.

Elements and Performance Criteria

Pulping and making of pulp products

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate how to chop and prepare water hyacinth for the pulping process.
- PC2.** Perform the chemical or mechanical pulping of water hyacinth using appropriate tools.
- PC3.** Measure the consistency and quality of water hyacinth pulp through simple tests.
- PC4.** Create basic pulp products, such as sheets, packaging materials, or molded items.
- PC5.** Demonstrate various steps to make different products
- PC6.** Demonstrate how to create Finished Products

Mould and Deckle Handmade paper making process

To be competent, the user/individual on the job must be able to:

- PC7.** Demonstrate Collection & processing of water hyacinth.
- PC8.** Make mould & deckle using appropriate tools.
- PC9.** Demonstrate Hand-made paper making process
- PC10.** Create Handmade paper products and work on their finishing
- PC11.** Inspect pulp products for defects, such as uneven texture or weak spots.

Product finishing, binding

To be competent, the user/individual on the job must be able to:

- PC12.** Explain the importance of finishing in enhancing the quality of pulp products made from water hyacinth.
- PC13.** Identify different binding techniques used in paper and pulp product manufacturing.
- PC14.** Describe the steps involved in the finishing process, including smoothing, trimming, and embossing.
- PC15.** Discuss the types of adhesives and materials used for binding pulp products.
- PC16.** Prepare pulp products from water hyacinth for the finishing process.

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- PC17.** Demonstrate smoothing techniques using hand tools to refine pulp surfaces.
- PC18.** Trim the edges of pulp products to ensure evenness and uniformity.
- PC19.** Show how to apply embossing techniques to add texture or decorative patterns.
- PC20.** Perform the process of stitching paper products using a needle and thread.
- PC21.** Test the durability of bound pulp products by applying pressure or water resistance tests.
- PC22.** Decorate pulp products with sustainable binding materials such as natural fibers or recycled threads.
- PC23.** Document the binding techniques used, noting adjustments for improving product quality.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamentals of pulp making from plant biomass.
- KU2.** Tools and machines used for chopping and pulping.
- KU3.** Properties of water hyacinth suitable for pulp.
- KU4.** Steps involved in converting pulp into paper sheets.
- KU5.** Types of paper products that can be made (sheets, packaging, etc.).
- KU6.** Basic quality parameters of handmade paper (thickness, smoothness, color).
- KU7.** Common defects in handmade paper and how to prevent them.
- KU8.** Finishing methods: drying, trimming, and embossing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Operate pulping tools and machines safely and efficiently.
- GS2.** Follow process instructions and maintain hygiene.
- GS3.** Document quantities of raw materials and finished pulp products.
- GS4.** Collaborate with peers in group paper-making activities.
- GS5.** Demonstrate creativity in product design and aesthetics.
- GS6.** Apply problem-solving skills when defects occur.
- GS7.** Communicate steps of the process to others effectively.
- GS8.** Show patience and accuracy in making uniform sheets.
- GS9.** Adapt to feedback for improving product quality.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pulping and making of pulp products</i>	-	12	-	-
PC1. Demonstrate how to chop and prepare water hyacinth for the pulping process.	-	2	-	-
PC2. Perform the chemical or mechanical pulping of water hyacinth using appropriate tools.	-	2	-	-
PC3. Measure the consistency and quality of water hyacinth pulp through simple tests.	-	2	-	-
PC4. Create basic pulp products, such as sheets, packaging materials, or molded items.	-	2	-	-
PC5. Demonstrate various steps to make different products	-	2	-	-
PC6. Demonstrate how to create Finished Products	-	2	-	-
<i>Mould and Deckle Handmade paper making process</i>	-	10	-	-
PC7. Demonstrate Collection & processing of water hyacinth.	-	2	-	-
PC8. Make mould & deckle using appropriate tools.	-	2	-	-
PC9. Demonstrate Hand-made paper making process	-	2	-	-
PC10. Create Handmade paper products and work on their finishing	-	2	-	-
PC11. Inspect pulp products for defects, such as uneven texture or weak spots.	-	2	-	-
<i>Product finishing, binding</i>	12	16	-	-
PC12. Explain the importance of finishing in enhancing the quality of pulp products made from water hyacinth.	3	-	-	-
PC13. Identify different binding techniques used in paper and pulp product manufacturing.	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Describe the steps involved in the finishing process, including smoothing, trimming, and embossing.	3	-	-	-
PC15. Discuss the types of adhesives and materials used for binding pulp products.	3	-	-	-
PC16. Prepare pulp products from water hyacinth for the finishing process.	-	2	-	-
PC17. Demonstrate smoothing techniques using hand tools to refine pulp surfaces.	-	2	-	-
PC18. Trim the edges of pulp products to ensure evenness and uniformity.	-	2	-	-
PC19. Show how to apply embossing techniques to add texture or decorative patterns.	-	2	-	-
PC20. Perform the process of stitching paper products using a needle and thread.	-	2	-	-
PC21. Test the durability of bound pulp products by applying pressure or water resistance tests.	-	2	-	-
PC22. Decorate pulp products with sustainable binding materials such as natural fibers or recycled threads.	-	2	-	-
PC23. Document the binding techniques used, noting adjustments for improving product quality.	-	2	-	-
NOS Total	12	38	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4119
NOS Name	Utilise water Hyacinth for Pulp Processing and Paper making
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4.0
Credits	1.0
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N4120: Utilise water Hyacinth for Weaving and handicraft making.

Description

This NOS enables individuals to convert dried and processed water hyacinth into attractive and functional handicrafts using traditional and modern weaving techniques. Learners gain practical skills in preparing raw material, weaving basic and complex patterns, assembling finished products, and applying decorative elements.

Scope

The scope covers the following :

- The scope covers the utilization of processed water hyacinth for creating woven and handcrafted products. It includes the identification, preparation, and treatment of water hyacinth fibers, understanding weaving techniques, and the creation of various handicraft items such as mats, baskets, and decorative articles.

Elements and Performance Criteria

Processing of Water Hyacinth for Weaving and handicraft making.

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the steps involved in processing water hyacinth for use in weaving and handicraft making.
- PC2.** Identify the different properties of water hyacinth that make it suitable for weaving.
- PC3.** Describe the preparation techniques for cleaning and drying water hyacinth before weaving.
- PC4.** Discuss the types of tools and equipment commonly used in weaving with water hyacinth.
- PC5.** Demonstrate cleaning techniques to remove impurities from water hyacinth fibres.
- PC6.** Perform drying methods to prepare water hyacinth for storage and future use.
- PC7.** Cut and organize water hyacinth fibres into manageable sizes for weaving.
- PC8.** Weave simple patterns using water hyacinth fibres, such as mats, baskets, or mats.
- PC9.** Practice basic weaving techniques such as over-under, plain weave, or twill.
- PC10.** Assemble multiple strips of water hyacinth into more complex woven products.
- PC11.** Inspect finished woven products for quality, ensuring tightness, consistency, and durability.
- PC12.** Decorate water hyacinth woven items using additional natural fibres, beads, or other embellishments.
- PC13.** Document the steps and outcomes of processing water hyacinth for weaving, noting areas for improvement.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basics of traditional and modern weaving methods.
- KU2.** Types of weaves: plain, twill, basket, etc.

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- KU3.** Preparation of water hyacinth fibers for weaving (cleaning, drying).
- KU4.** Required tools (loom, frame, scissors) and their use.
- KU5.** Techniques for product shaping and joining.
- KU6.** Finishing methods for woven items.
- KU7.** Use of natural dyes or eco-friendly additives.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate design ideas clearly with team or trainer.
- GS2.** Read and follow pattern instructions.
- GS3.** Work patiently to maintain accuracy and aesthetics.
- GS4.** Collaborate in weaving teams for larger products.
- GS5.** Maintain records of items created and materials used.
- GS6.** Practice cleanliness and safe tool handling.
- GS7.** Improve product design based on customer feedback.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Processing of Water Hyacinth for Weaving and handicraft making.</i>	20	30	-	-
PC1. Explain the steps involved in processing water hyacinth for use in weaving and handicraft making.	5	-	-	-
PC2. Identify the different properties of water hyacinth that make it suitable for weaving.	5	-	-	-
PC3. Describe the preparation techniques for cleaning and drying water hyacinth before weaving.	5	-	-	-
PC4. Discuss the types of tools and equipment commonly used in weaving with water hyacinth.	5	-	-	-
PC5. Demonstrate cleaning techniques to remove impurities from water hyacinth fibres.	-	3	-	-
PC6. Perform drying methods to prepare water hyacinth for storage and future use.	-	3	-	-
PC7. Cut and organize water hyacinth fibres into manageable sizes for weaving.	-	3	-	-
PC8. Weave simple patterns using water hyacinth fibres, such as mats, baskets, or mats.	-	3	-	-
PC9. Practice basic weaving techniques such as over-under, plain weave, or twill.	-	3	-	-
PC10. Assemble multiple strips of water hyacinth into more complex woven products.	-	3	-	-
PC11. Inspect finished woven products for quality, ensuring tightness, consistency, and durability.	-	4	-	-
PC12. Decorate water hyacinth woven items using additional natural fibres, beads, or other embellishments.	-	4	-	-
PC13. Document the steps and outcomes of processing water hyacinth for weaving, noting areas for improvement.	-	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4120
NOS Name	Utilise water Hyacinth for Weaving and handicraft making.
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4.0
Credits	1.0
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N4121: Utilise water Hyacinth for Mushroom Cultivation

Description

This unit enables individuals to utilize water hyacinth an invasive aquatic weed as a sustainable substrate for cultivating edible mushrooms. Learners are trained in collecting, cleaning, and processing water hyacinth for bed preparation, selecting appropriate mushroom species, and maintaining ideal growth conditions.

Scope

The scope covers the following :

- The Scope covers the use of water hyacinth as a substrate for mushroom cultivation. It includes knowledge and skills related to the preparation of mushroom beds using processed water hyacinth, identification of suitable mushroom species, cultivation techniques, and post-harvest practices such as packaging and value addition.

Elements and Performance Criteria

Processing of Water Hyacinth for bed preparation/substrates for mushroom cultivation. Identify different species of Mushroom. Packaging of Mushrooms and Value addition

To be competent, the user/individual on the job must be able to:

- | | |
|--------------|---|
| PC1. | Explain the importance of using water hyacinth as a substrate for mushroom cultivation. |
| PC2. | Describe the processing steps required to prepare water hyacinth for mushroom bed creation. |
| PC3. | Discuss the nutritional benefits of water hyacinth as a substrate for mushroom growth. |
| PC4. | Identify different types of mushrooms that can be cultivated on water hyacinth substrates. |
| PC5. | Explain the factors that influence the success of mushroom cultivation using water hyacinth. |
| PC6. | Discuss the packaging techniques that help maintain the freshness and quality of harvested mushrooms. |
| PC7. | Collect water hyacinth suitable for use as a substrate for mushroom cultivation. |
| PC8. | Prepare water hyacinth by cleaning, drying, and shredding it into smaller pieces. |
| PC9. | Inoculate the substrate with mushroom spawn, ensuring proper spacing and coverage. |
| PC10. | Monitor the moisture levels in the water hyacinth substrate to support healthy mushroom growth. |
| PC11. | Harvest mushrooms at the appropriate stage of development based on visual and growth indicators. |
| PC12. | Package harvested mushrooms using breathable containers to maintain freshness and prevent spoilage. |

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- PC13.** Label packaged mushrooms clearly with production date and other relevant details.
- PC14.** Evaluate the quality of mushrooms by assessing size, color, texture, and aroma.
- PC15.** Record observations during each step of mushroom cultivation and packaging, identifying areas for improvement.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Role of water hyacinth as a mushroom substrate.
- KU2.** Different species of mushrooms suitable for this medium.
- KU3.** Step-by-step bed preparation and inoculation process.
- KU4.** Conditions for mushroom growth (humidity, temperature, pH).
- KU5.** Indicators for healthy and mature mushrooms.
- KU6.** Value addition options using mushrooms (drying, pickling).
- KU7.** Packaging standards for fresh and processed mushrooms.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Maintain growth and harvesting logs.
- GS2.** Follow hygiene and sanitation protocols strictly.
- GS3.** Communicate with buyers on mushroom care and use.
- GS4.** Work with team to prepare large mushroom beds.
- GS5.** Practice safe handling and packaging.
- GS6.** Promote products with local branding and story-telling.
- GS7.** Record and analyze customer feedback.
- GS8.** Apply innovation in designing mushroom-based products.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Processing of Water Hyacinth for bed preparation/substrates for mushroom cultivation. Identify different species of Mushroom. Packaging of Mushrooms and Value addition</i>	25	25	-	-
PC1. Explain the importance of using water hyacinth as a substrate for mushroom cultivation.	4	-	-	-
PC2. Describe the processing steps required to prepare water hyacinth for mushroom bed creation.	4	-	-	-
PC3. Discuss the nutritional benefits of water hyacinth as a substrate for mushroom growth.	4	-	-	-
PC4. Identify different types of mushrooms that can be cultivated on water hyacinth substrates.	4	-	-	-
PC5. Explain the factors that influence the success of mushroom cultivation using water hyacinth.	4	-	-	-
PC6. Discuss the packaging techniques that help maintain the freshness and quality of harvested mushrooms.	5	-	-	-
PC7. Collect water hyacinth suitable for use as a substrate for mushroom cultivation.	-	2	-	-
PC8. Prepare water hyacinth by cleaning, drying, and shredding it into smaller pieces.	-	2	-	-
PC9. Inoculate the substrate with mushroom spawn, ensuring proper spacing and coverage.	-	2	-	-
PC10. Monitor the moisture levels in the water hyacinth substrate to support healthy mushroom growth.	-	3	-	-
PC11. Harvest mushrooms at the appropriate stage of development based on visual and growth indicators.	-	3	-	-
PC12. Package harvested mushrooms using breathable containers to maintain freshness and prevent spoilage.	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Label packaged mushrooms clearly with production date and other relevant details.	-	3	-	-
PC14. Evaluate the quality of mushrooms by assessing size, color, texture, and aroma.	-	3	-	-
PC15. Record observations during each step of mushroom cultivation and packaging, identifying areas for improvement.	-	4	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4121
NOS Name	Utilise water Hyacinth for Mushroom Cultivation
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4.0
Credits	1.0
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N4122: Utilise water Hyacinth for Briquette production.

Description

This unit enables individuals to utilize water hyacinth an invasive aquatic weed as a raw material for making biomass briquettes. Learners are trained in drying and shredding the plant, mixing it with other organic materials if required, and operating manual or mechanical briquetting machines to form uniform briquettes. The NOS also includes quality checks, safe handling, and packaging of the finished product. This process supports renewable energy initiatives, reduces dependency on firewood and fossil fuels, and promotes green micro-enterprises in rural and semi-urban communities.

Scope

The scope covers the following :

- The Scope covers the conversion of water hyacinth into biomass briquettes used as an alternative fuel source. It includes the collection and preparation of water hyacinth, understanding its fuel properties, the operation of briquetting machines, and the production, drying, testing, and packaging of briquettes.

Elements and Performance Criteria

Briquette production, raw material and machines involved.

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the process of briquette production and its benefits as an alternative fuel.
- PC2.** Describe the different raw materials that can be used for briquette production, including water hyacinth.
- PC3.** Discuss the chemical composition and properties of water hyacinth that make it suitable for briquette production.
- PC4.** Identify the types of machines and equipment commonly used in the briquetting process.
- PC5.** Illustrate the importance of moisture control in raw materials during briquette making.
- PC6.** Explain the different types of briquettes (e.g., biomass briquettes) and their applications.
- PC7.** Collect water hyacinth and other suitable organic materials for briquette production.
- PC8.** Prepare the raw materials by cleaning and drying them to remove excess moisture.
- PC9.** Shred water hyacinth into smaller pieces to facilitate easier blending and binding.
- PC10.** Operate the briquetting machine to press the raw materials into uniform briquettes.
- PC11.** Monitor the pressing process to ensure the correct density and size of the briquettes.
- PC12.** Inspect the quality of the briquettes by assessing density, shape, and durability.
- PC13.** Test the combustion properties of the briquettes to evaluate their fuel efficiency.
- PC14.** Package the briquettes in appropriate packaging to prevent moisture absorption and damage.
- PC15.** Record observations during each step of briquette production, noting areas for improvement.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Concept of biomass briquettes and their energy benefits.
- KU2.** Types of briquetting machines (manual, semi-auto, electric).
- KU3.** Role of water hyacinth as a binder or filler.
- KU4.** Drying, grinding, and size reduction processes.
- KU5.** Importance of moisture control during briquette formation.
- KU6.** Shape, size, and strength standards for briquettes.
- KU7.** Storage and packaging practices to avoid degradation.
- KU8.** Economic benefits of biomass fuels in rural areas.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Operate machinery with safety and accuracy.
- GS2.** Follow SOPs for moisture measurement and binding.
- GS3.** Maintain daily records of raw material and output.
- GS4.** Work in teams for material collection and mixing.
- GS5.** Demonstrate proper lifting and handling of tools.
- GS6.** Identify problems with shape or density and take corrective steps.
- GS7.** Communicate benefits of briquettes to community users.
- GS8.** Ensure cleanliness and fire safety in the production area.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Briquette production, raw material and machines involved.</i>	23	27	-	-
PC1. Explain the process of briquette production and its benefits as an alternative fuel.	4	-	-	-
PC2. Describe the different raw materials that can be used for briquette production, including water hyacinth.	4	-	-	-
PC3. Discuss the chemical composition and properties of water hyacinth that make it suitable for briquette production.	4	-	-	-
PC4. Identify the types of machines and equipment commonly used in the briquetting process.	3	-	-	-
PC5. Illustrate the importance of moisture control in raw materials during briquette making.	4	-	-	-
PC6. Explain the different types of briquettes (e.g., biomass briquettes) and their applications.	4	-	-	-
PC7. Collect water hyacinth and other suitable organic materials for briquette production.	-	3	-	-
PC8. Prepare the raw materials by cleaning and drying them to remove excess moisture.	-	3	-	-
PC9. Shred water hyacinth into smaller pieces to facilitate easier blending and binding.	-	3	-	-
PC10. Operate the briquetting machine to press the raw materials into uniform briquettes.	-	3	-	-
PC11. Monitor the pressing process to ensure the correct density and size of the briquettes.	-	3	-	-
PC12. Inspect the quality of the briquettes by assessing density, shape, and durability.	-	3	-	-
PC13. Test the combustion properties of the briquettes to evaluate their fuel efficiency.	-	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Package the briquettes in appropriate packaging to prevent moisture absorption and damage.	-	3	-	-
PC15. Record observations during each step of briquette production, noting areas for improvement.	-	3	-	-
NOS Total	23	27	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4122
NOS Name	Utilise water Hyacinth for Briquette production.
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4.0
Credits	1.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N4123: Utilise water Hyacinth for Biogas production.

Description

This unit enables individuals to convert water hyacinth into biogas, a clean and renewable source of energy. Learners gain practical skills in collecting and shredding water hyacinth, preparing it as feedstock, installing biogas units, and managing the digestion process to ensure optimal gas output. It also includes safety measures during operation and maintenance, as well as the handling of slurry by-products.

Scope

The scope covers the following :

- The Scope covers the use of water hyacinth as a feedstock for biogas generation through anaerobic digestion. It includes the collection and preparation of raw material, understanding biogas plant components and installation, and maintaining safe and efficient operations.

Elements and Performance Criteria

Biogas production, installation and precautions.

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the process of biogas production and its benefits as a renewable energy source.
- PC2.** Describe the different types of biogas digesters and their components.
- PC3.** Discuss the chemical and biological principles behind the anaerobic digestion process.
- PC4.** Identify the types of organic materials that can be used for biogas production, including water hyacinth.
- PC5.** Explain the steps involved in the installation of a biogas plant.
- PC6.** Illustrate the importance of maintaining proper pH, temperature, and retention time in biogas production.
- PC7.** Discuss the safety precautions necessary during biogas production and installation.
- PC8.** Collect organic materials, such as water hyacinth, suitable for biogas production.
- PC9.** Prepare feedstocks by cleaning, shredding, and mixing them to ensure proper digestion.
- PC10.** Install the biogas digester according to manufacturer guidelines and environmental conditions.
- PC11.** Operate the biogas plant by feeding the prepared organic materials into the digester.
- PC12.** Monitor the temperature, pH, and pressure inside the biogas digester to optimize conditions for digestion.
- PC13.** Test the biogas production process by measuring gas output using appropriate equipment.
- PC14.** Maintain the biogas digester by regularly inspecting and cleaning the components.
- PC15.** Apply safety precautions while handling biogas and biogas plant components.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Anaerobic digestion process for biogas generation.
- KU2.** Types of digesters and their working principles.
- KU3.** Materials suitable for feedstock (water hyacinth, dung, kitchen waste).
- KU4.** Process of loading, mixing, and stirring feed materials.
- KU5.** Safety protocols in handling gas leaks or digester failures.
- KU6.** Biogas usage for lighting, cooking, and power.
- KU7.** Maintenance steps and frequency for plant health.
- KU8.** Community-scale biogas as a sustainable energy solution.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Record gas production and feed input regularly.
- GS2.** Work in a team to set up or monitor digesters.
- GS3.** Communicate clearly about safety and maintenance.
- GS4.** Demonstrate safe handling of slurry and equipment.
- GS5.** Use digital tools for data logging, if available.
- GS6.** Prepare charts/posters on biogas for local promotion.
- GS7.** Interpret readings from pressure gauges and thermometers.
- GS8.** Manage waste slurry as compost for agriculture.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Biogas production, installation and precautions.</i>	20	30	-	-
PC1. Explain the process of biogas production and its benefits as a renewable energy source.	3	-	-	-
PC2. Describe the different types of biogas digesters and their components.	3	-	-	-
PC3. Discuss the chemical and biological principles behind the anaerobic digestion process.	3	-	-	-
PC4. Identify the types of organic materials that can be used for biogas production, including water hyacinth.	3	-	-	-
PC5. Explain the steps involved in the installation of a biogas plant.	3	-	-	-
PC6. Illustrate the importance of maintaining proper pH, temperature, and retention time in biogas production.	3	-	-	-
PC7. Discuss the safety precautions necessary during biogas production and installation.	2	-	-	-
PC8. Collect organic materials, such as water hyacinth, suitable for biogas production.	-	3	-	-
PC9. Prepare feedstocks by cleaning, shredding, and mixing them to ensure proper digestion.	-	3	-	-
PC10. Install the biogas digester according to manufacturer guidelines and environmental conditions.	-	4	-	-
PC11. Operate the biogas plant by feeding the prepared organic materials into the digester.	-	4	-	-
PC12. Monitor the temperature, pH, and pressure inside the biogas digester to optimize conditions for digestion.	-	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Test the biogas production process by measuring gas output using appropriate equipment.	-	4	-	-
PC14. Maintain the biogas digester by regularly inspecting and cleaning the components.	-	4	-	-
PC15. Apply safety precautions while handling biogas and biogas plant components.	-	4	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4123
NOS Name	Utilise water Hyacinth for Biogas production.
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4.0
Credits	1.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N4124: Finishing and Marketing of Products

Description

This unit enables individuals to refine and market a range of products made from water hyacinth such as handmade paper, handicrafts, briquettes, or mushroom-related goods. Learners are trained in product finishing techniques like smoothing, trimming, stitching, and assembling to improve aesthetics and usability. They also learn to package products attractively, develop branding materials, and apply marketing principles including the 4Ps (Product, Price, Place, Promotion).

Scope

The scope covers the following :

- The Scope covers the essential skills and knowledge required for the finishing, packaging, branding, and marketing of water hyacinth-based value-added products. It includes techniques to enhance product appearance and durability, as well as methods to promote and sell products through both traditional and digital channels.

Elements and Performance Criteria

Finishing and Marketing of the products

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the importance of finishing techniques in enhancing the quality and marketability of products made from water hyacinth.
- PC2.** Describe the various finishing methods used to improve the appearance, texture, and durability of water hyacinth products.
- PC3.** Discuss the key factors that influence the market demand for handmade products made from water hyacinth.
- PC4.** Explain 4Ps of marketing, managing the customer.
- PC5.** Identify different marketing channels and platforms suitable for promoting water hyacinth products.
- PC6.** Illustrate techniques for packaging that protect water hyacinth products and enhance their appeal to buyers.
- PC7.** Explain the role of social media and digital marketing in promoting handmade water hyacinth products.
- PC8.** Discuss ways to build customer relationships and gain repeat business for water hyacinth products.
- PC9.** Clean and prepare water hyacinth products for the finishing process.
- PC10.** Apply smoothing techniques using hand tools to enhance the surface finish of water hyacinth items.
- PC11.** Trim edges of products to ensure uniformity and neat appearance.
- PC12.** Assemble components into completed products with precision and care.
- PC13.** Package water hyacinth products using appropriate materials to ensure protection and presentation.

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- PC14.** Label products clearly with relevant information, such as brand name, material used, and care instructions.
- PC15.** Market products by creating attractive product displays for local fairs, exhibitions, or online platforms.
- PC16.** Promote products using social media, including creating engaging posts and responding to customer inquiries.
- PC17.** Record sales data and customer feedback to assess the effectiveness of marketing strategies and make improvements.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Finishing methods to enhance look, strength, and usability.
- KU2.** Importance of trimming, smoothing, coloring, or polishing.
- KU3.** Basic packaging principles to protect and attract.
- KU4.** Labelling guidelines including eco-labels and product info.
- KU5.** Concepts of branding, pricing, and positioning.
- KU6.** Understanding the 4Ps of marketing.
- KU7.** Online and offline marketing channels (fairs, social media).
- KU8.** Customer relationship management and repeat sales.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate product features confidently to customers.
- GS2.** Handle sales transactions and maintain records.
- GS3.** Maintain neat finishing and defect-free packaging.
- GS4.** Follow up with customers for feedback or referrals.
- GS5.** Prepare product labels using eco-friendly materials.
- GS6.** Manage product stock efficiently and update inventory.
- GS7.** Present in exhibitions or community events.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Finishing and Marketing of the products</i>	20	30	-	-
PC1. Explain the importance of finishing techniques in enhancing the quality and marketability of products made from water hyacinth.	2	-	-	-
PC2. Describe the various finishing methods used to improve the appearance, texture, and durability of water hyacinth products.	2	-	-	-
PC3. Discuss the key factors that influence the market demand for handmade products made from water hyacinth.	2	-	-	-
PC4. Explain 4Ps of marketing, managing the customer.	2	-	-	-
PC5. Identify different marketing channels and platforms suitable for promoting water hyacinth products.	3	-	-	-
PC6. Illustrate techniques for packaging that protect water hyacinth products and enhance their appeal to buyers.	3	-	-	-
PC7. Explain the role of social media and digital marketing in promoting handmade water hyacinth products.	3	-	-	-
PC8. Discuss ways to build customer relationships and gain repeat business for water hyacinth products.	3	-	-	-
PC9. Clean and prepare water hyacinth products for the finishing process.	-	3	-	-
PC10. Apply smoothing techniques using hand tools to enhance the surface finish of water hyacinth items.	-	3	-	-
PC11. Trim edges of products to ensure uniformity and neat appearance.	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Assemble components into completed products with precision and care.	-	3	-	-
PC13. Package water hyacinth products using appropriate materials to ensure protection and presentation.	-	3	-	-
PC14. Label products clearly with relevant information, such as brand name, material used, and care instructions.	-	3	-	-
PC15. Market products by creating attractive product displays for local fairs, exhibitions, or online platforms.	-	4	-	-
PC16. Promote products using social media, including creating engaging posts and responding to customer inquiries.	-	4	-	-
PC17. Record sales data and customer feedback to assess the effectiveness of marketing strategies and make improvements.	-	4	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4124
NOS Name	Finishing and Marketing of Products
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4.0
Credits	1.0
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4118.Introduce course collection, cleaning and processing of water hyacinth.	20	30	0	0	50	16
SGJ/N4119.Utilise water Hyacinth for Pulp Processing and Paper making	12	38	0	0	50	12
SGJ/N4120.Utilise water Hyacinth for Weaving and handicraft making.	20	30	0	0	50	12
SGJ/N4121.Utilise water Hyacinth for Mushroom Cultivation	25	25	0	0	50	12
SGJ/N4122.Utilise water Hyacinth for Briquette production.	23	27	0	0	50	12
SGJ/N4123.Utilise water Hyacinth for Biogas production.	20	30	0	0	50	12
SGJ/N4124.Finishing and Marketing of Products	20	30	0	0	50	12
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	12
Total	160	240	-	-	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.